

CSL351 : COMPUTER NETWORKS

Assignment - 3

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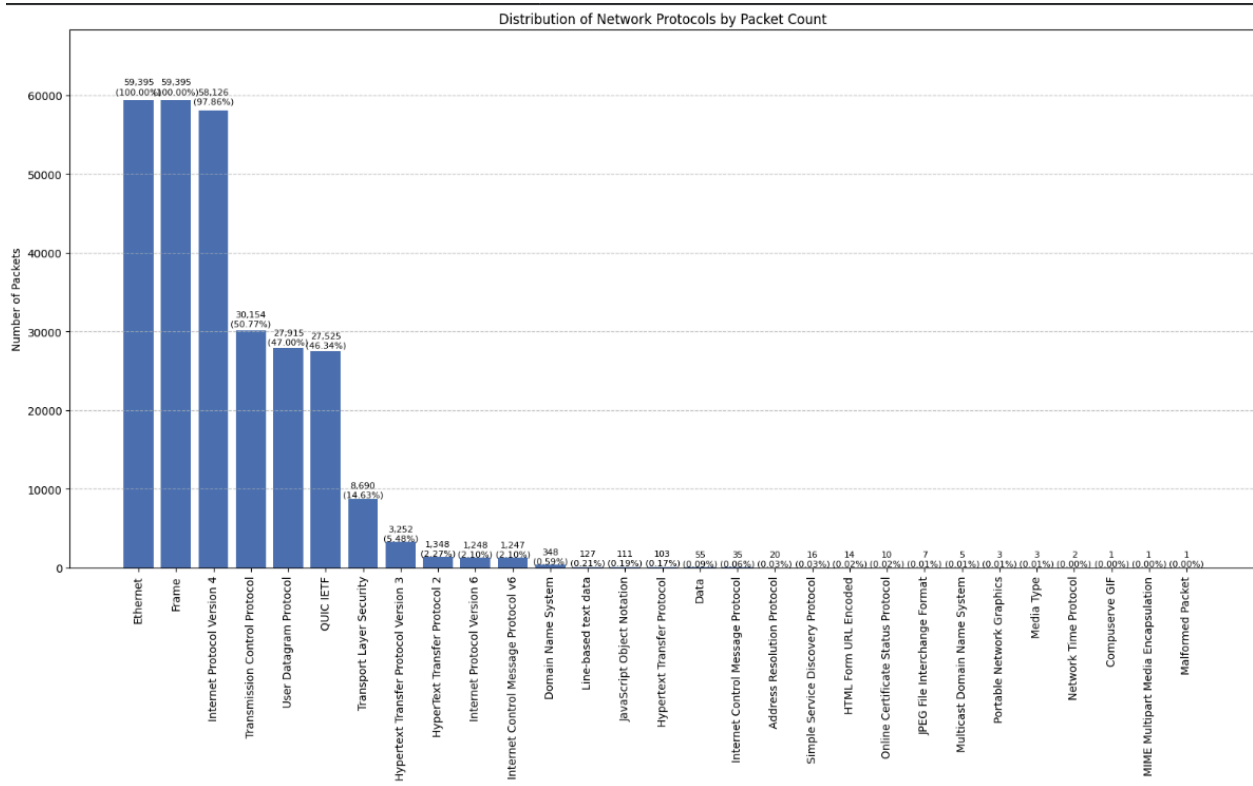
Link for Pcap File :

<https://drive.google.com/file/d/1mXHXGbwhEeyhbCGqRPbgxdlQ3v3rpki/vi?usp=sharing>

Que 1.

Total of 59395 packets were captured.

Measurement	Captured	Displayed	Marked
Packets	59395	59395 (100.0%)	—
Time span, s	725.858	725.858	—
Average pps	81.8	81.8	—
Average packet size, B	1530	1530	—
Bytes	90880073	90880073 (100.0%)	0
Average bytes/s	125 k	125 k	—
Average bits/s	1,001 k	1,001 k	—



Que 2.

HTTP3 is the most frequent protocol observed for a total count of 3252 packets.

Que 3.

No, wireshark also captures packets of activities not performed by the user. Like the 20 packets of the Address Resolution Protocol (ARP) ,2 packets of Network Time Protocol (NTP) which are used by the Operating System for network discovery , automated protocols etc.

Pcap file filtered for some protocols which is sent by OS.

No.	Time	Source	Destination	Protocol	Length	Info
4089	35.607248	Cisco_89:22:19	Intel_4a:52:4a	ARP	60	Who has 10.50.0.178? Tell 0.0.0.0
4090	35.607276	Intel_4a:52:4a	Cisco_89:22:19	ARP	42	10.50.0.178 is at 8c:e9:ee:4a:52:4a
4246	73.148733	Intel_4a:52:4a	Cisco_af:3e:02	ARP	42	Who has 10.50.0.1? Tell 10.50.0.178
4247	73.152700	Cisco_af:3e:02	Intel_4a:52:4a	ARP	60	10.50.0.1 is at 34:1b:2d:af:3e:02
9921	113.052105	Cisco_0d:69:ff	Intel_4a:52:4a	ARP	60	Who has 10.50.0.178? Tell 10.50.0.5
9922	113.052131	Intel_4a:52:4a	Cisco_0d:69:ff	ARP	42	10.50.0.178 is at 8c:e9:ee:4a:52:4a
11709	187.836658	Intel_4a:52:4a	Cisco_af:3e:02	ARP	42	Who has 10.50.0.1? Tell 10.50.0.178
11710	187.839296	Cisco_af:3e:02	Intel_4a:52:4a	ARP	60	10.50.0.1 is at 34:1b:2d:af:3e:02
30721	341.331672	Cisco_89:22:19	Intel_4a:52:4a	ARP	60	Who has 10.50.0.178? Tell 0.0.0.0
30722	341.331696	Intel_4a:52:4a	Cisco_89:22:19	ARP	42	10.50.0.178 is at 8c:e9:ee:4a:52:4a
50953	428.888878	Cisco_0d:69:ff	Intel_4a:52:4a	ARP	60	Who has 10.50.0.178? Tell 10.50.0.5
50954	428.888909	Intel_4a:52:4a	Cisco_0d:69:ff	ARP	42	10.50.0.178 is at 8c:e9:ee:4a:52:4a
59051	578.492719	Intel_4a:52:4a	Cisco_af:3e:02	ARP	42	Who has 10.50.0.1? Tell 10.50.0.178
59052	578.517265	Cisco_af:3e:02	Intel_4a:52:4a	ARP	60	10.50.0.1 is at 34:1b:2d:af:3e:02
59218	643.935035	Cisco_89:22:19	Intel_4a:52:4a	ARP	60	Who has 10.50.0.178? Tell 0.0.0.0
59219	643.935052	Intel_4a:52:4a	Cisco_89:22:19	ARP	42	10.50.0.178 is at 8c:e9:ee:4a:52:4a
59243	649.660729	Intel_4a:52:4a	Cisco_af:3e:02	ARP	42	Who has 10.50.0.1? Tell 10.50.0.178
59244	649.665068	Cisco_af:3e:02	Intel_4a:52:4a	ARP	60	10.50.0.1 is at 34:1b:2d:af:3e:02
59298	675.260725	Intel_4a:52:4a	Cisco_af:3e:02	ARP	42	Who has 10.50.0.1? Tell 10.50.0.178
59300	675.283598	Cisco_af:3e:02	Intel_4a:52:4a	ARP	60	10.50.0.1 is at 34:1b:2d:af:3e:02

No.	Time	Source	Destination	Protocol	Length	Info
3351	23.393757	10.50.0.178	185.125.190.57	NTP	90	NTP Version 4, client
3352	23.652917	185.125.190.57	10.50.0.178	NTP	90	NTP Version 4, server

Que 4.

The image shows the connection for downloading the pdf, between the client and server.

Filter used : **http.request.uri contains "handbook"**

Statistics			
Measurement	Captured	Displayed	Marked
Packets	59395	239 (0.4%)	—
Time span, s	725.858	60.335	—
Average pps	81.8	4.0	—
Average packet size, B	1530	4927	—
Bytes	90880073	1177533 (1.3%)	0
Average bytes/s	125 k	19 k	—
Average bits/s	1,001 k	156 k	—

Total packets captured were 239.

Screenshot showing the client server

Time	10.50.0.178	192.168.10.115	Comment
28.525495	47878	47878 → 443 [SYN] Seq=0 Win=64240 Len=0 ...	TCP: 47878 → 443 [SYN] Seq=0 Win=64240 Len=0 ...
28.527770	47878	443 → 47878 [SYN, ACK] Seq=0 Ack=1 Win=2...	TCP: 443 → 47878 [SYN, ACK] Seq=0 Ack=1 Win=2...
28.527781	47878	47878 → 443 [ACK] Seq=1 Ack=1 Win=...	TCP: 47878 → 443 [ACK] Seq=1 Ack=1 Win=...
28.527977	47878	Client Hello (SNI=www.iitbhilai.ac.in)	TLSv1.2: Client Hello (SNI=www.iitbhilai.ac.in)
28.529644	47878	443 → 47878 [ACK] Seq=1 Ack=1239 Win=31...	TCP: 443 → 47878 [ACK] Seq=1 Ack=1239 Win=31...
28.530055	47878	443 → 47878 [ACK] Seq=1 Ack=2002 Win=33...	TCP: 443 → 47878 [ACK] Seq=1 Ack=2002 Win=33...
28.530055	47878	Server Hello, Change Cipher Spec, Fini...	TLSv1.2: Server Hello, Change Cipher Spec, Finished
28.530078	47878	47878 → 443 [ACK] Seq=2002 Ack=13...	TCP: 47878 → 443 [ACK] Seq=2002 Ack=13 Win=...
28.530216	47878	Change Cipher Spec, Finished	TLSv1.2: Change Cipher Spec, Finished
28.530393	47878	GET /index.php?pid=css_style HTTP/1.1	HTTP: GET /index.php?pid=css_style HTTP/1.1
28.533967	47878	443 → 47878 [ACK] Seq=138 Ack=2725 Win=...	TCP: 443 → 47878 [ACK] Seq=138 Ack=2725 Win=...
28.967132	47878	[TLS segment of a reassembled PDU]	TLSv1.2: [TLS segment of a reassembled PDU]
28.967265	47878	47878 → 443 [ACK] Seq=2725 Ack=5090 Win=...	TCP: 47878 → 443 [ACK] Seq=2725 Ack=5090 Win=...
28.967298	47878	443 → 47878 [ACK] Seq=5090 Ack=2725 Win=...	TCP: 443 → 47878 [ACK] Seq=5090 Ack=2725 Win=...
28.967344	47878	47878 → 443 [ACK] Seq=2725 Ack=8804 Win=...	TCP: 47878 → 443 [ACK] Seq=2725 Ack=8804 Win=...
28.967716	47878	443 → 47878 [ACK] Seq=8804 Ack=2725 Win=...	TCP: 443 → 47878 [ACK] Seq=8804 Ack=2725 Win=...
28.969460	47878	443 → 47878 [ACK] Seq=11280 Ack=2725 Wi...	TCP: 443 → 47878 [ACK] Seq=11280 Ack=2725 Wi...
28.970113	47878	47878 → 443 [ACK] Seq=2725 Ack=12518 Wi...	TCP: 47878 → 443 [ACK] Seq=2725 Ack=12518 Wi...
28.970939	47878	HTTP/1.1 200 OK (text/css)	TLSv1.2: HTTP/1.1 200 OK (text/css)
28.971158	47878	47878 → 443 [ACK] Seq=2725 Ack=19...	TCP: 47878 → 443 [ACK] Seq=2725 Ack=19897 Wi...
28.979650	47878	[TCP Spurious Retransmission] 443 → ...	TCP: [TCP Spurious Retransmission] 443 → 47878 [...]
28.979669	47878	[TCP Dup ACK 3615#1] 47878 → 443 [ACK] S...	TCP: [TCP Dup ACK 3615#1] 47878 → 443 [ACK] S...
28.985486	47878	GET /index.php?pid=js_effi_cryptojs H...	HTTP: GET /index.php?pid=js_effi_cryptojs HTTP/...
29.026197	47878	443 → 47878 [ACK] Seq=19897 Ack=3390 Wi...	TCP: 443 → 47878 [ACK] Seq=19897 Ack=3390 Wi...
29.32042	47878	[TLS segment of a reassembled PDU]	TLSv1.2: [TLS segment of a reassembled PDU]
29.392167	47878	47878 → 443 [ACK] Seq=3390 Ack=35991 Wi...	TCP: 47878 → 443 [ACK] Seq=3390 Ack=35991 Wi...
29.394281	47878	[TLS segment of a reassembled PDU]	TLSv1.2: [TLS segment of a reassembled PDU]
29.394303	47878	47878 → 443 [ACK] Seq=3390 Ack=40943 Wi...	TCP: 47878 → 443 [ACK] Seq=3390 Ack=40943 Wi...
29.394664	47878	443 → 47878 [ACK] Seq=40943 Ack=3390 Wi...	TCP: 443 → 47878 [ACK] Seq=40943 Ack=3390 Wi...
29.394675	47878	47878 → 443 [ACK] Seq=3390 Ack=49609 Wi...	TCP: 47878 → 443 [ACK] Seq=3390 Ack=49609 Wi...

Screenshot showing flow of packets :



Que 5.

Ping uses the ICMP protocol. So to identify the packets of “ping tesla.com” we look for the ICMP Protocol with IP as ‘2.18.53.207’ in the wireshark.

No.	Time	Source	Destination	Protocol	Length	Info
58846	519.123118	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=1/256, ttl=64 (no response found!)
58850	520.124944	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=2/512, ttl=64 (no response found!)
58853	521.148956	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=3/768, ttl=64 (no response found!)
58855	522.172675	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=4/1024, ttl=64 (no response found!)
58858	523.196885	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=5/1280, ttl=64 (no response found!)
58861	524.221002	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=6/1536, ttl=64 (no response found!)
58865	525.244793	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=7/1792, ttl=64 (no response found!)
58869	526.268896	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=8/2048, ttl=64 (no response found!)
58873	527.292963	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=9/2304, ttl=64 (no response found!)
58876	528.316992	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=10/2560, ttl=64 (no response found!)
58879	529.340903	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=11/2816, ttl=64 (no response found!)
58881	530.364946	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=12/3072, ttl=64 (no response found!)
58886	531.388951	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=13/3328, ttl=64 (no response found!)
58894	532.413878	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=14/3584, ttl=64 (no response found!)
58898	533.436850	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=15/3840, ttl=64 (no response found!)
58904	534.460901	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=16/4096, ttl=64 (no response found!)
58910	535.484927	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=17/4352, ttl=64 (no response found!)
58914	536.509037	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=18/4608, ttl=64 (no response found!)
58916	537.532917	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=19/4864, ttl=64 (no response found!)
58923	538.556948	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=20/5120, ttl=64 (no response found!)
58927	539.580966	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=21/5376, ttl=64 (no response found!)
58929	540.604975	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=22/5632, ttl=64 (no response found!)
58938	541.628899	10.50.0.178	2.18.53.207	ICMP	1073	Echo (ping) request id=0x3c0a, seq=23/5888, ttl=64 (no response found!)

We verify the terminal output as we can see from the terminal image 100% packets were lost and in the image of the wireshark filtered with ICMP protocol we can see ‘no response found!’.

```
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VV-FX507VV: ~  
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VV-FX507VV:~$ ping tesla.com -s 2511  
PING tesla.com (2.18.53.207) 2511(2539) bytes of data.  
^C  
--- tesla.com ping statistics ---  
23 packets transmitted, 0 received, 100% packet loss, time 22505ms  
  
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VV-FX507VV:~$ traceroute iitbhilai.ac.in  
traceroute to iitbhilai.ac.in (192.168.10.115), 30 hops max, 60 byte packets  
1 _gateway (10.50.0.1) 8.836 ms 9.097 ms 9.085 ms  
2 10.200.10.14 (10.200.10.14) 1.835 ms 1.822 ms 2.139 ms  
3 192.168.10.115 (192.168.10.115) 2.473 ms * 2.447 ms  
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VV-FX507VV:~$ 12340370 this is my roll no  
.
```

Que 6.

Saving files in .pcap is useful because we can know the exact timing and contents of the packet which is used to analyse and troubleshoot the packets later in offline mode.

Que 7.

Filter expression used : 'dns || http'

A total of 451 packets were shown after this filter was enabled.

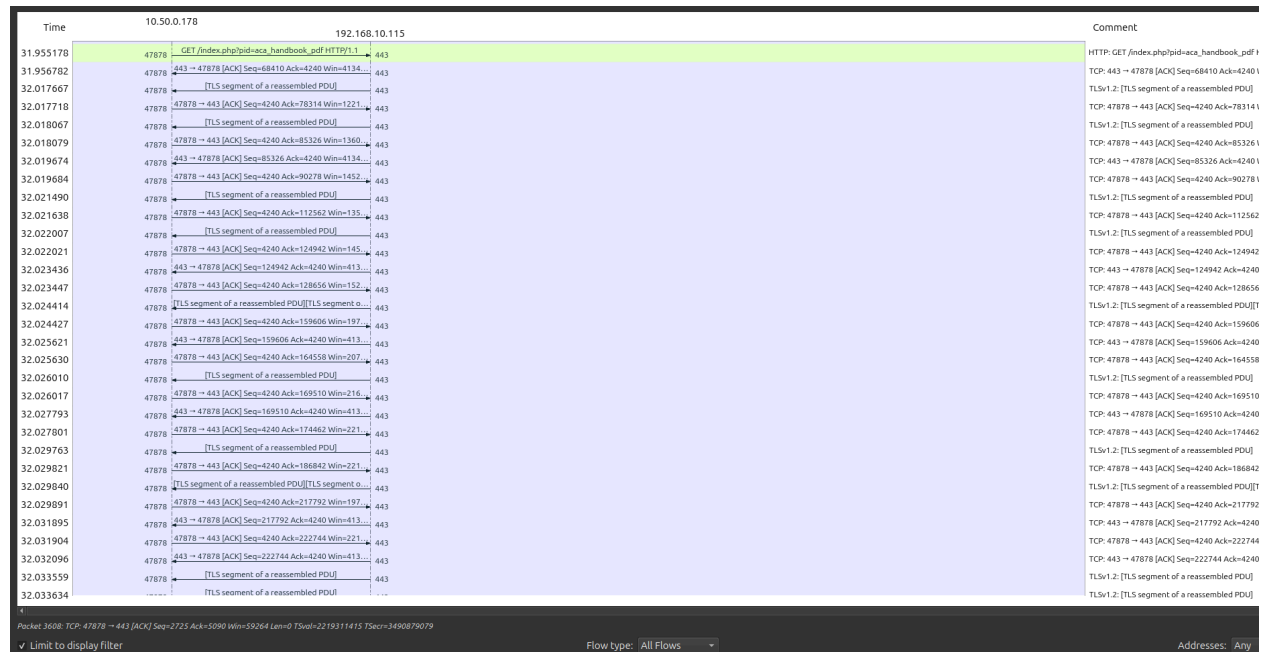
dns http					
No.	Time	Source	Destination	Protocol	Length Info
197	16.053237	192.168.10.72	10.50.0.178	DNS	130 Standard query response 0xe248 HTTPS www.iitbhlai.ac.in SOA dns2.iitbhlai.ac.in OPT
200	16.054578	192.168.10.72	10.50.0.178	DNS	141 Standard query response 0x855a A www.iitbhlai.ac.in A 192.168.10.115 NS dns2.iitbhlai.ac.in A 192.168.10.72 OPT
251	16.138724	10.50.0.178	192.168.10.115	HTTP	768 GET / HTTP/1.1
295	16.314596	192.168.10.115	10.50.0.178	TLSv1.2	6944 HTTP/1.1 200 OK (text/html)
304	16.338784	10.50.0.178	192.168.10.72	DNS	94 Standard query 0xa80e HTTPS safebrowsing.google.com OPT
308	16.358444	10.50.0.178	192.168.10.72	DNS	84 Standard query 0xa83d A safebrowsing.google.com OPT

Statistics			
Measurement	Captured	Displayed	Marked
Packets	59395	451 (0.8%)	—
Time span, s	725.858	602.759	—
Average pps	81.8	0.7	—
Average packet size, B	1530	847	—
Bytes	90880073	381995 (0.4%)	0
Average bytes/s	125 k	633	—
Average bits/s	1,001 k	5,069	—

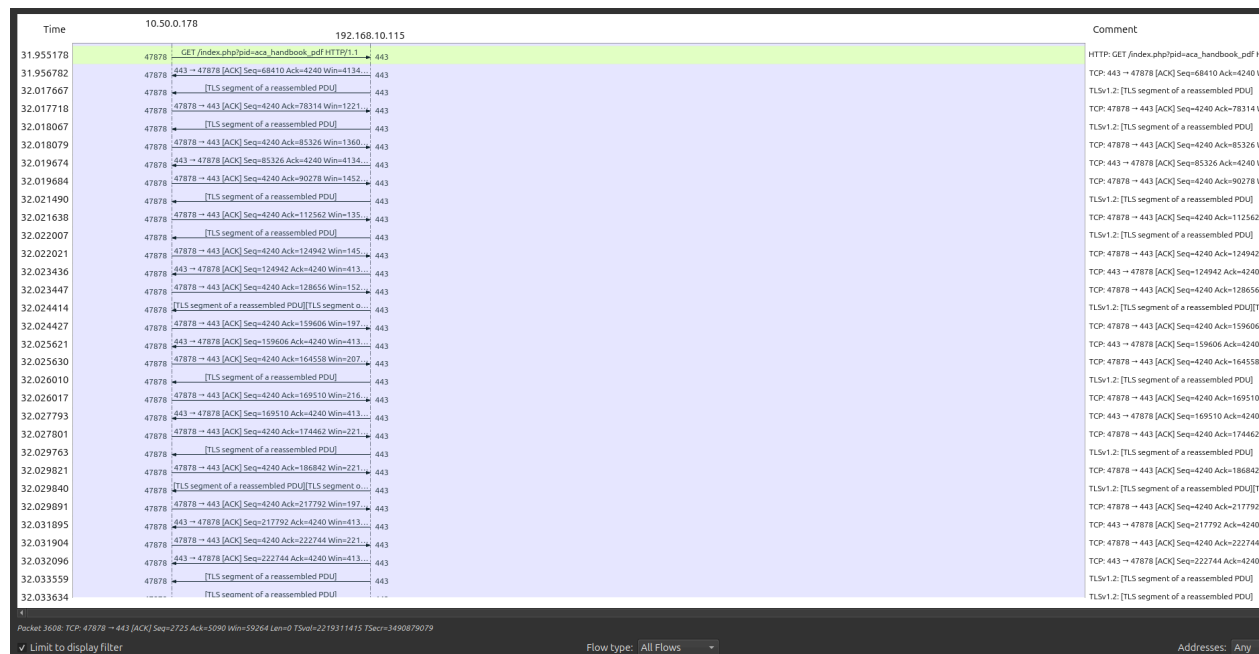
PART 2 : STATISTICS

Que 1

Flow graph while downloading the handbook



Que 2



The process of downloading works as follows :

The file downloading sequence starts with the client (your laptop) and the iitbhlai server establishing a TCP connection using the three-way handshake (SYN , ACK etc). After the connection is set up a TLS handshake is performed to encrypt the entire session. The client requests the handbook.pdf via an HTTP GET request which shows up as ‘Application Data’ in the flow graph. Finally the server transmits the file content in numerous encrypted segments and the client sends periodic ACK packets to confirm successful receipt until the download is complete.

Que 3

Metric	Value
Total Packets Captured	59395
Total Number of HTTP/HTTPS connections established	53
Average Throughput of streaming videos	235.16 Kbps
How many DNS queries were resolved?	174
Average throughput of QUIC Protocols	235.16 Kbps
Total Number of SMTP packets captured	47
Average Packet Length	1530.10
Average Packet Length of TCP Traffic	2332.39
Average time spent on TLS Message Exchange	0.104 sec
How many packets are retransmitted?	26

Que 4

Time	10.50.0.178	2.18.53.207	10.200.10.14	192.168.10.115	10.50.0.1	Comment
519.123118	Echo (ping) request id=0x3cda, seq=17256, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=17256, .
520.124944	Echo (ping) request id=0x3cda, seq=22512, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=22512, .
521.148956	Echo (ping) request id=0x3cda, seq=37768, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=37768, .
522.172675	Echo (ping) request id=0x3cda, seq=471024, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=471024, .
523.196885	Echo (ping) request id=0x3cda, seq=571280, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=571280, .
524.221002	Echo (ping) request id=0x3cda, seq=671536, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=671536, .
525.244793	Echo (ping) request id=0x3cda, seq=771792, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=771792, .
526.268896	Echo (ping) request id=0x3cda, seq=872048, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=872048, .
527.292963	Echo (ping) request id=0x3cda, seq=972304, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=972304, .
528.316992	Echo (ping) request id=0x3cda, seq=1072560, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=107256, .
529.340903	Echo (ping) request id=0x3cda, seq=1172816, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=117281, .
530.364946	Echo (ping) request id=0x3cda, seq=1273072, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=127307, .
531.388951	Echo (ping) request id=0x3cda, seq=1373328, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=137332, .
532.413878	Echo (ping) request id=0x3cda, seq=1473584, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=147358, .
533.436850	Echo (ping) request id=0x3cda, seq=1573840, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=157384, .
534.460901	Echo (ping) request id=0x3cda, seq=1674096, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=167409, .
535.484927	Echo (ping) request id=0x3cda, seq=1774352, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=177435, .
536.509037	Echo (ping) request id=0x3cda, seq=1874608, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=187460, .
537.532917	Echo (ping) request id=0x3cda, seq=1974864, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=197486, .
538.556948	Echo (ping) request id=0x3cda, seq=2075120, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=207512, .
539.580966	Echo (ping) request id=0x3cda, seq=2175376, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=217537, .
540.604975	Echo (ping) request id=0x3cda, seq=2275632, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=227563, .
541.628889	Echo (ping) request id=0x3cda, seq=2375888, ttl=64 [in]					ICMP: Echo (ping) request id=0x3cda, seq=237588, .
552.292217	33437 Time-to-live exceeded (Time to live exceeded in transit)		45465			ICMP: Time-to-live exceeded (Time to live exceed..
552.292217	33438 Time-to-live exceeded (Time to live exceeded in transit)		56194			ICMP: Time-to-live exceeded (Time to live exceed..
552.292546	33439 Time-to-live exceeded (Time to live exceeded in transit)		44703			ICMP: Time-to-live exceeded (Time to live exceed..
552.292893	33444 Destination unreachable (Port unreachable)			50128		ICMP: Destination unreachable (Port unreachable)
552.292893	33442 Destination unreachable (Port unreachable)			40821		ICMP: Destination unreachable (Port unreachable)
552.292893	33443 Destination unreachable (Port unreachable)			34307		ICMP: Destination unreachable (Port unreachable)
552.292893	33447 Destination unreachable (Port unreachable)			55465		ICMP: Destination unreachable (Port unreachable)

Traceroute works as follows :

Traceroute determines the path to a destination by sending multiple packets each with a time to live (TTL) value that increases by one starting at 1. The first router receives the packet with a ttl of 1 reduces it to 0 discards it and sends back an ICMP Time Exceeded error thus identifying itself. This process is repeated. With each subsequent increase in the ttl value (2 3 etc.) the next router in the path identifies itself by returning a similar error. This continues until the packet successfully reaches the final destination which responds with a standard reply instead of an error thereby mapping the complete network route.

Que 5.

Yes we can identify the protocols used for sending mail using thunderbird as well as gmail (web) ,

For the first method we have the following flow graph (thunderbird) :



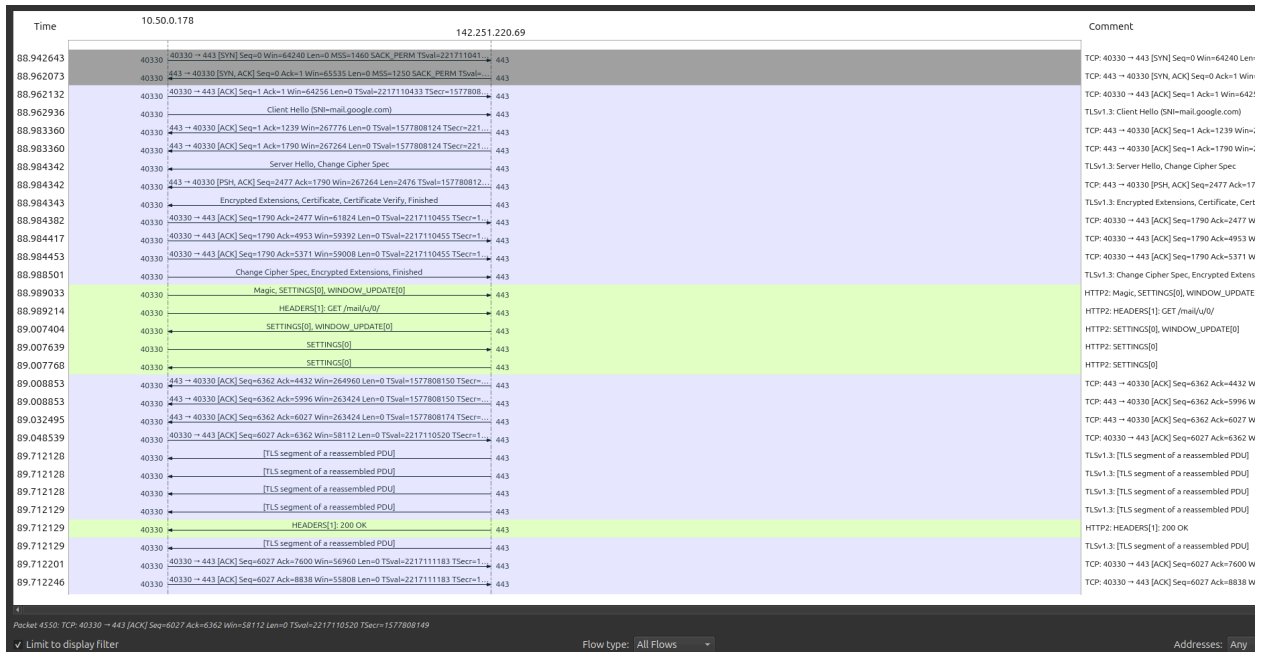
The sequential order of protocols :

TCP - for connection

TLS - for handshake

SMTP - for email commands inside TLS tunnel

For the second method we have the following flow graph (gmail) :



The sequential order of protocols :

DNS - to find the domain the of gmail.com

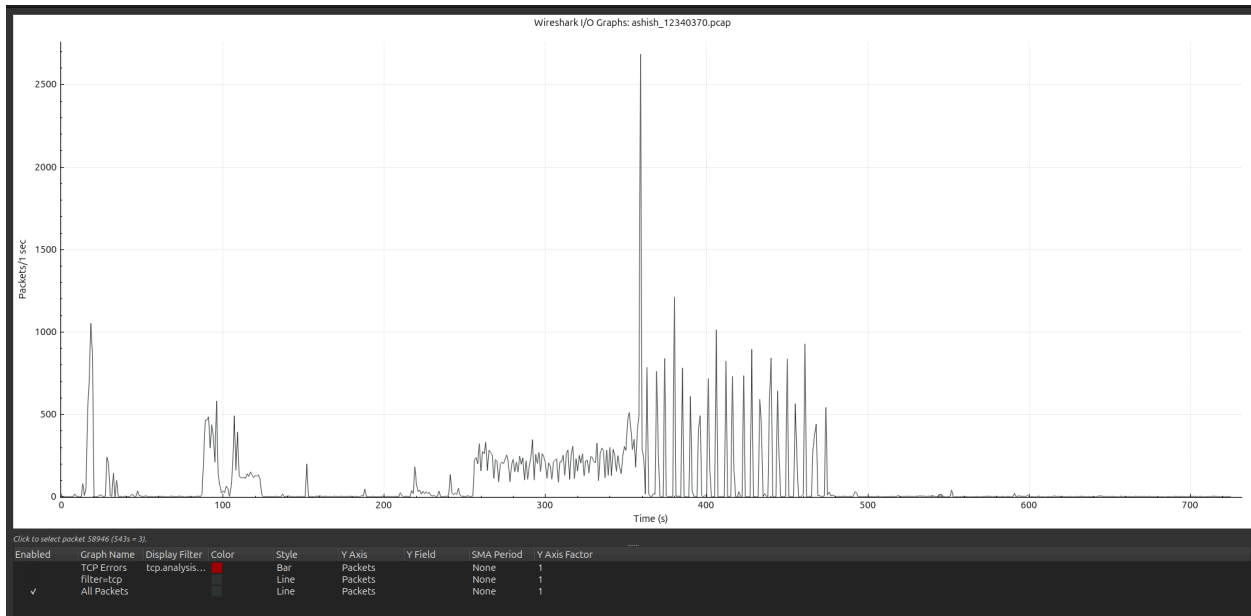
TCP - to build connection with handshake using SYN,ACK etc

TLS - to secure the connection using encryption

HTTP/3 - to send post ,get request and the application data packet.

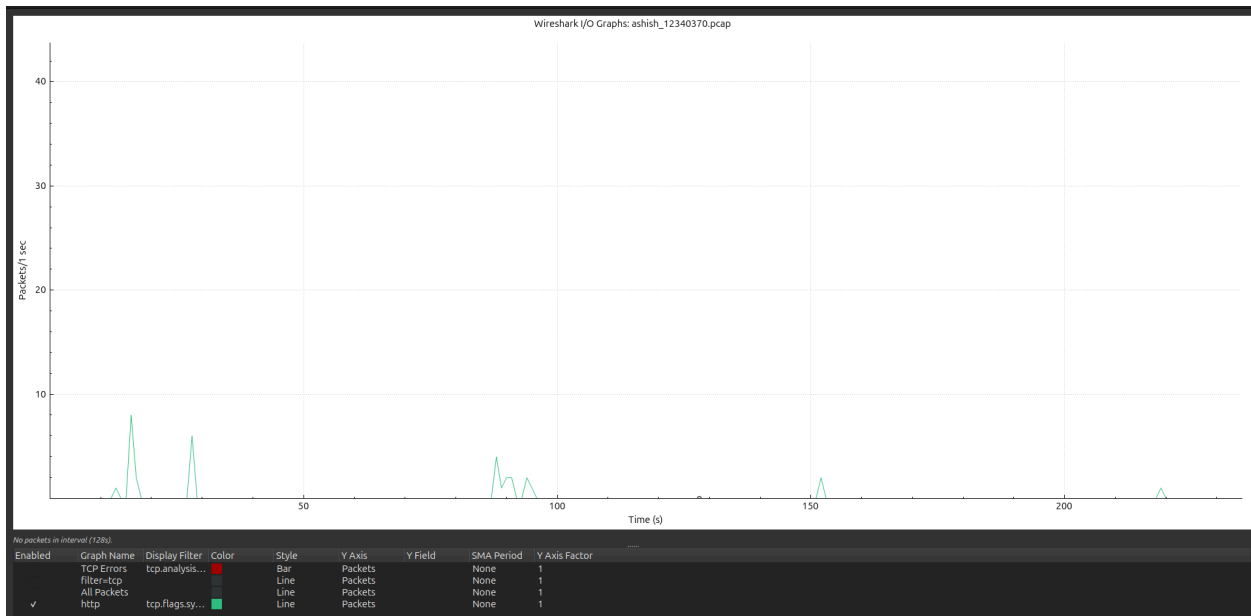
Que 6 .

a. For complete packets captured



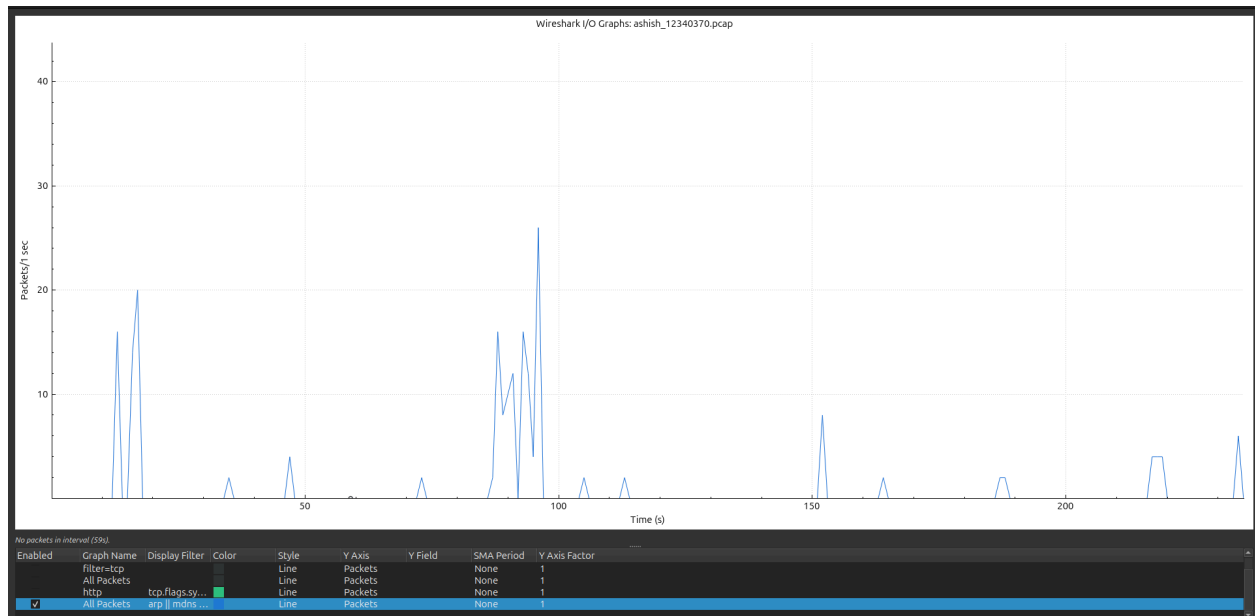
b. For HTTP/HTTPS traffic

Filter used : `tcp.flags.syn == 1 && tcp.flags.ack == 0 && (tcp.port == 80 || tcp.port == 443 )`



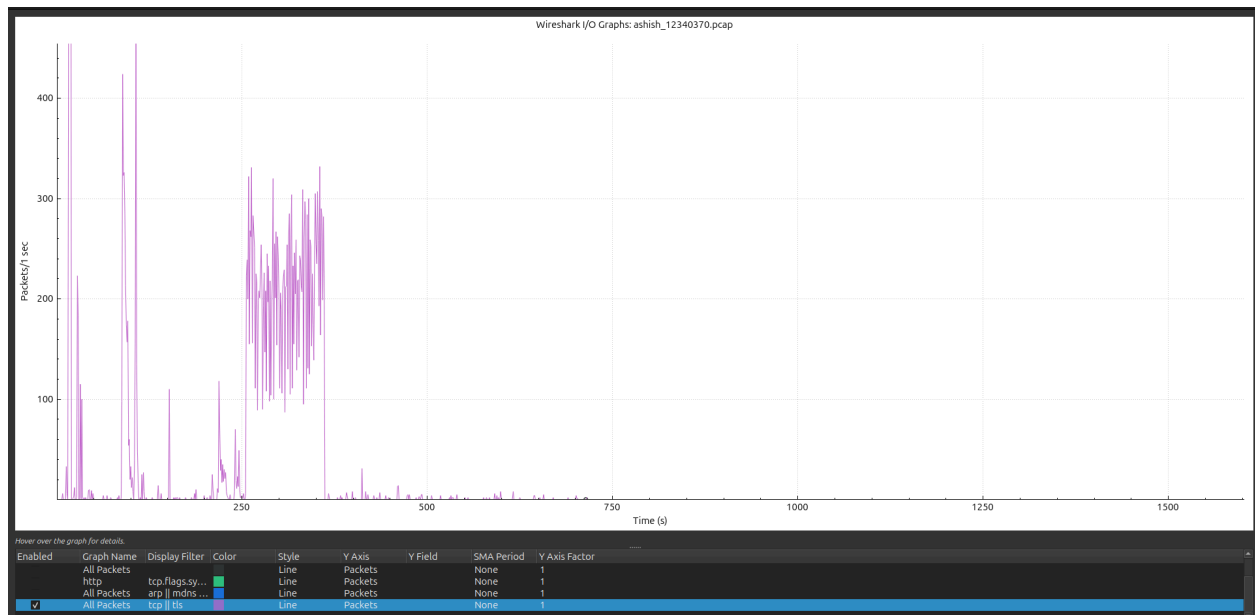
c. For All the ARP, DNS/MDNS, and ICMP Queries

Filter used : arp || mdns || dns || icmp



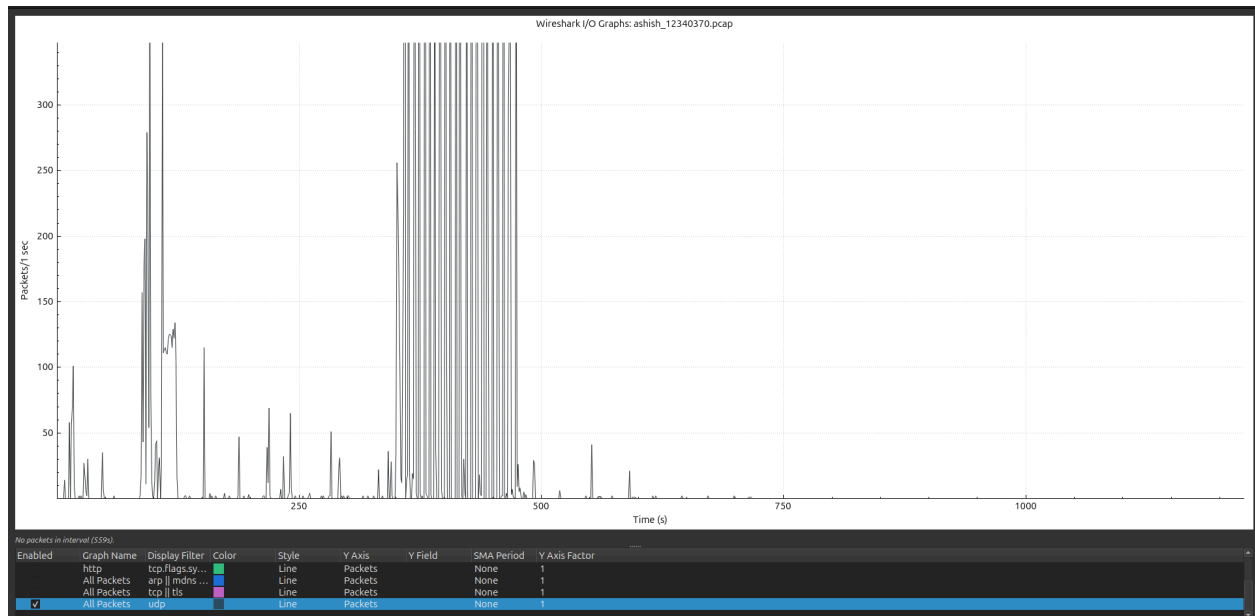
d. For TCP/TLS Protocols

Filter used : tcp || tls



e. For UDP protocol

Filter used : udp



Ques 7

a. For youtube.com

Filter used : ip.addr == 180.149.60.12

Time	10.50.0.178	180.149.60.12	Comment
358.957610	49419	Initial, DCID=6de1acf6de24e031, PKN: 1, PING, CRYPTO	QUIC: Initial, DCID=6de1acf6de24e031, PKN: 1, P...
358.957630	49419	Initial, DCID=6de1acf6de24e031, PKN: 2, CRYPTO, C...	QUIC: Initial, DCID=6de1acf6de24e031, PKN: 2, C...
358.989549	49419	Initial, SCID=e0e1acf6de24e031, PKN: 1, ACK	QUIC: Initial, SCID=e0e1acf6de24e031, PKN: 1, ACK
358.992707	49419	Initial, SCID=e0e1acf6de24e031, PKN: 2, ACK, PADDING	QUIC: Initial, SCID=e0e1acf6de24e031, PKN: 2, AC...
358.992707	49419	Handshake, SCID=e0e1acf6de24e031, PKN: 4, CRYPTO	QUIC: Handshake, SCID=e0e1acf6de24e031, PKN:...
358.992707	49419	Handshake, SCID=e0e1acf6de24e031, PKN: 5, CRYPTO	QUIC: Handshake, SCID=e0e1acf6de24e031, PKN:...
358.992707	49419	Protected Payload (KPO), PKN: 8, STREAM(3), SETTINGS	HTTP/3: Protected Payload (KPO), PKN: 8, STREAM...
358.992707	49419	Handshake, SCID=e0e1acf6de24e031, PKN: 5, CRYPTO	QUIC: Handshake, SCID=e0e1acf6de24e031, PKN:...
358.992882	49419	Handshake, DCID=e0e1acf6de24e031, PKN: 3, ACK	QUIC: Handshake, DCID=e0e1acf6de24e031, PKN:...
358.992905	49419	Handshake, DCID=e0e1acf6de24e031, PKN: 4, ACK	QUIC: Handshake, DCID=e0e1acf6de24e031, PKN:...
358.993011	49419	Handshake, DCID=e0e1acf6de24e031, PKN: 5, ACK	QUIC: Handshake, DCID=e0e1acf6de24e031, PKN:...
358.994639	49419	Handshake, DCID=e0e1acf6de24e031, PKN: 6, CRYPTO	QUIC: Handshake, DCID=e0e1acf6de24e031, PKN:...
358.994713	49419	Protected Payload (KPO), DCID=e0e1acf6de24e031	HTTP/3: Protected Payload (KPO), DCID=e0e1acf6...
358.995113	49419	Protected Payload (KPO), DCID=e0e1acf6de24e031	HTTP/3: Protected Payload (KPO), DCID=e0e1acf6...
358.995124	49419	Protected Payload (KPO), DCID=e0e1acf6de24e031	HTTP/3: Protected Payload (KPO), DCID=e0e1acf6...
358.995129	49419	Protected Payload (KPO), DCID=e0e1acf6de24e031	HTTP/3: Protected Payload (KPO), DCID=e0e1acf6...
359.027160	49419	Protected Payload (KPO), PKN: 9, CRYPTO	QUIC: Protected Payload (KPO), PKN: 9, CRYPTO
359.027160	49419	Protected Payload (KPO), PKN: 10, DONE, NT, NC	QUIC: Protected Payload (KPO), PKN: 10, DONE, N...
359.027160	49419	Protected Payload (KPO), PKN: 11, ACK	QUIC: Protected Payload (KPO), PKN: 11, ACK
359.027160	49419	Protected Payload (KPO), PKN: 12, ACK, STREAM(7)	HTTP/3: Protected Payload (KPO), PKN: 12, ACK, S...
359.027325	49419	Protected Payload (KPO), DCID=e0e1acf6de24e031	QUIC: Protected Payload (KPO), DCID=e0e1acf6de...
359.027911	49419	Protected Payload (KPO), PKN: 13, STREAM(1), STR...	HTTP/3: Protected Payload (KPO), PKN: 13, STRA...
359.028038	49419	Protected Payload (KPO), DCID=e0e1acf6de24e031	HTTP/3: Protected Payload (KPO), DCID=e0e1acf6...
359.084911	49419	Protected Payload (KPO), PKN: 14, ACK	QUIC: Protected Payload (KPO), PKN: 14, ACK
359.334218	49419	Protected Payload (KPO), PKN: 15, STREAM(0)	QUIC: Protected Payload (KPO), PKN: 15, STREAM(0)
359.334218	49419	Protected Payload (KPO), PKN: 16, STREAM(0)	QUIC: Protected Payload (KPO), PKN: 16, STREAM(0)
359.334218	49419	Protected Payload (KPO), PKN: 17, STREAM(0)	QUIC: Protected Payload (KPO), PKN: 17, STREAM(0)
359.334218	49419	Protected Payload (KPO), PKN: 18, STREAM(0)	QUIC: Protected Payload (KPO), PKN: 18, STREAM(0)
359.334218	49419	Protected Payload (KPO), PKN: 19, STREAM(0)	QUIC: Protected Payload (KPO), PKN: 19, STREAM(0)
359.334218	49419	Protected Payload (KPO), PKN: 20, STREAM(0)	QUIC: Protected Payload (KPO), PKN: 20, STREAM(0)

Packet 36036: HTTP/3: Protected Payload (KPO), DCID=e0e1acf6de24e031, PKN: 7, STREAM(2), SETTINGS

Limit to display filter

Flow type: All Flows

Addresses: Any

b. For Accessing iitbhillai.ac.in

Filter used : ip.addr == 192.168.10.115

Time	10.50.0.178	192.168.10.72	192.168.10.115	Comment
16.013239	35421	Standard query 0x6248 HTTPS www.iitbhillai.ac.in OPT	53	DNS: Standard query 0x6248 HTTPS www.iitbhillai...
16.013294	53040	Standard query 0x855a A www.iitbhillai.ac.in OPT	53	DNS: Standard query 0x855a A www.iitbhillai.ac.in...
16.053237	35421	Standard query response 0x6248 HTTPS www.iitbhillai.ac.in OPT	53	DNS: Standard query response 0x6248 HTTPS www...
16.054578	53040	Standard query response 0x855a A www.iitbhillai.ac.in OPT	53	DNS: Standard query response 0x855a A www.iitb...
16.138724	35616	GET / HTTP/1.1	443	HTTP: GET / HTTP/1.1
16.330933	35616	GET /index.php?ipid=css_bootstrapmin HTTP/1.1	443	HTTP: GET /index.php?ipid=css_bootstrapmin HT...
16.335188	35630	GET /index.php?ipid=css_style HTTP/1.1	443	HTTP: GET /index.php?ipid=css_style HTTP/1.1
16.340530	35644	GET /index.php?ipid=css_bootstrap_select HTTP/1.1	443	HTTP: GET /index.php?ipid=css_bootstrap_select ...
16.340901	35658	GET /index.php?ipid=css_fontawesomemin HTTP/1.1	443	HTTP: GET /index.php?ipid=css_fontawesomemin ...
16.341285	35666	GET /index.php?ipid=js_search HTTP/1.1	443	HTTP: GET /index.php?ipid=js_search HTTP/1.1
16.343036	35678	GET /index.php?ipid=js_querymin HTTP/1.1	443	HTTP: GET /index.php?ipid=js_querymin HTTP/1.1
16.404754	35616	GET /index.php?ipid=js_bootstrap_select HTTP/1.1	443	HTTP: GET /index.php?ipid=js_bootstrap_select HT...
16.467277	35644	GET /index.php?ipid=img_logo HTTP/1.1	443	HTTP: GET /index.php?ipid=img_logo HTTP/1.1
16.523184	35630	GET /index.php?ipid=img_gpc2026 HTTP/1.1	443	HTTP: GET /index.php?ipid=img_gpc2026 HTTP/1.1
16.574507	35644	GET /index.php?ipid=fltrb_convocation HTTP/1.1	443	HTTP: GET /index.php?ipid=fltrb_convocation HT...
16.694986	35644	GET /index.php?ipid=independence_day_79th HTTP/1.1	443	HTTP: GET /index.php?ipid=independence_day_79...
16.855452	35658	GET /index.php?ipid=yoga_2025 HTTP/1.1	443	HTTP: GET /index.php?ipid=yoga_2025 HTTP/1.1
16.855657	35666	GET /index.php?ipid=js_bootstrapmin HTTP/1.1	443	HTTP: GET /index.php?ipid=js_bootstrapmin HTTP/...
16.855690	35616	GET /index.php?ipid=js_effi_cryptojs HTTP/1.1	443	HTTP: GET /index.php?ipid=js_effi_cryptojs HTTP/...
16.855713	35678	GET /index.php?ipid=js_effi_cryptojs_hmacsha256 HTTP/1.1	443	HTTP: GET /index.php?ipid=js_effi_cryptojs_hmacs...
16.968174	35644	GET /index.php?ipid=img_transparent HTTP/1.1	443	HTTP: GET /index.php?ipid=img_transparent HT...
17.060125	35630	GET /index.php?ipid=js_effi_cryptojs_encbase64 HTTP/1.1	443	HTTP: GET /index.php?ipid=js_effi_cryptojs_encba...
17.097419	35616	GET /index.php?ipid=js_effi_serviceutility HTTP/1.1	443	HTTP: GET /index.php?ipid=js_effi_serviceutility H...
17.150911	35630	GET /index.php?ipid=WorldEnvironmentDay2025 HTTP/1.1	443	HTTP: GET /index.php?ipid=WorldEnvironmentDay...
17.201871	35616	GET /index.php?ipid=archaeology_archives_museum HTTP/1.1	443	HTTP: GET /index.php?ipid=archaeology_archives...
17.250718	35678	GET /index.php?ipid=img_republicday HTTP/1.1	443	HTTP: GET /index.php?ipid=img_republicday HTTP/...
17.476837	35658	GET /index.php?ipid=cdcl_launch HTTP/1.1	443	HTTP: GET /index.php?ipid=cdcl_launch HTTP/1.1
17.515686	35666	GET /index.php?ipid=Technology_Prioritization_Workshop HTTP/1.1	443	HTTP: GET /index.php?ipid=Technology_Prioritizat...
18.041503	35630	GET /index.php?ipid=interac_24_1 HTTP/1.1	443	HTTP: GET /index.php?ipid=interac_24_1 HTTP/1.1
18.168227	35644	GET /index.php?ipid=independence_2024 HTTP/1.1	443	HTTP: GET /index.php?ipid=independence_2024 H...
Packet 543: HTTP: GET /index.php?ipid=js_effi_cryptojs_hmacsha256 HTTP/1.1				
✓ Limit to display filter				
Flow type: All Flows				
Addresses: Any				

c. DNS Query

Filter used : dns

Time	10.50.0.178	192.168.10.72	Comment
13.630920	56856	Standard query 0x87d2 HTTPS android.clients.google.com	DNS: Standard query 0x87d2 HTTPS android.clients.google.com
13.631005	37983	Standard query 0x3864 A android.clients.google.com	DNS: Standard query 0x3864 A android.clients.google.com
13.635705	37983	Standard query response 0x3864 A android.clients.google.com	DNS: Standard query response 0x3864 A android.clients.google.com
13.635719	56856	Standard query response 0x87d2 HTTPS android.clients.google.com	DNS: Standard query response 0x87d2 HTTPS android.clients.google.com
13.637025	47562	Standard query 0xdcd5 HTTPS android.l.google.com	DNS: Standard query 0xdcd5 HTTPS android.l.google.com
13.639221	47562	Standard query response 0xdcd5 HTTPS android.l.google.com	DNS: Standard query response 0xdcd5 HTTPS android.l.google.com
13.756667	37853	Standard query 0xa63f HTTPS beacons.gcp.gvt2.com	DNS: Standard query 0xa63f HTTPS beacons.gcp.gvt2.com
13.756908	56955	Standard query 0xc1f4 A beacons.gcp.gvt2.com OPT	DNS: Standard query 0xc1f4 A beacons.gcp.gvt2.com OPT
13.758574	37853	Standard query response 0xa63f HTTPS beacons.gcp.gvt2.com	DNS: Standard query response 0xa63f HTTPS beacons.gcp.gvt2.com
13.758945	57049	Standard query 0xd6fc HTTPS beacons-handoff.gcp.gvt2.com	DNS: Standard query 0xd6fc HTTPS beacons-handoff.gcp.gvt2.com
13.758994	56955	Standard query response 0xc1f4 A beacons.gcp.gvt2.com	DNS: Standard query response 0xc1f4 A beacons.gcp.gvt2.com
13.760844	57049	Standard query response 0xd6fc HTTPS beacons-handoff.gcp.gvt2.com	DNS: Standard query response 0xd6fc HTTPS beacons-handoff.gcp.gvt2.com
13.875942	40295	Standard query 0x2739 HTTPS e2c71.gcp.gvt2.com	DNS: Standard query 0x2739 HTTPS e2c71.gcp.gvt2.com
13.876131	51417	Standard query 0xd61 A e2c71.gcp.gvt2.com OPT	DNS: Standard query 0xd61 A e2c71.gcp.gvt2.com OPT
13.928660	51417	Standard query response 0xd61 A e2c71.gcp.gvt2.com	DNS: Standard query response 0xd61 A e2c71.gcp.gvt2.com
13.928836	40295	Standard query response 0x2739 HTTPS e2c71.gcp.gvt2.com	DNS: Standard query response 0x2739 HTTPS e2c71.gcp.gvt2.com
16.008257	37128	Standard query 0x55e7 HTTPS optimizationguide-pa.googleapis.com	DNS: Standard query 0x55e7 HTTPS optimizationguide-pa.googleapis.com
16.008416	40592	Standard query 0x4d3c A optimizationguide-pa.googleapis.com	DNS: Standard query 0x4d3c A optimizationguide-pa.googleapis.com
16.013239	35421	Standard query 0x6248 HTTPS www.iitbhillai.ac.in OPT	DNS: Standard query 0x6248 HTTPS www.iitbhillai.ac.in OPT
16.013294	53040	Standard query 0x855a A www.iitbhillai.ac.in OPT	DNS: Standard query 0x855a A www.iitbhillai.ac.in OPT
16.051290	37128	Standard query response 0x55e7 HTTPS optimizationguide-pa.googleapis.com	DNS: Standard query response 0x55e7 HTTPS optimizationguide-pa.googleapis.com
16.051293	40592	Standard query response 0x4d3c A optimizationguide-pa.googleapis.com	DNS: Standard query response 0x4d3c A optimizationguide-pa.googleapis.com
16.053237	35421	Standard query response 0x6248 HTTPS www.iitbhillai.ac.in OPT	DNS: Standard query response 0x6248 HTTPS www.iitbhillai.ac.in OPT
16.054578	53040	Standard query response 0x855a A www.iitbhillai.ac.in OPT	DNS: Standard query response 0x855a A www.iitbhillai.ac.in OPT
16.330794	38656	Standard query 0xa80e HTTPS safefbrowsing.google.com	DNS: Standard query 0xa80e HTTPS safefbrowsing.google.com
16.330914	58284	Standard query 0x9a2d A safefbrowsing.google.com	DNS: Standard query 0x9a2d A safefbrowsing.google.com
16.333957	58284	Standard query response 0x9a2d A safefbrowsing.google.com	DNS: Standard query response 0x9a2d A safefbrowsing.google.com
16.376443	38656	Standard query response 0xa80e HTTPS safefbrowsing.google.com	DNS: Standard query response 0xa80e HTTPS safefbrowsing.google.com
16.376701	34318	Standard query 0xf1ec HTTPS sb.l.google.com OPT	DNS: Standard query 0xf1ec HTTPS sb.l.google.com OPT
16.379004	34318	Standard query response 0xf1ec HTTPS sb.l.google.com	DNS: Standard query response 0xf1ec HTTPS sb.l.google.com
Packet 191: DNS: Standard query 0x855a A www.iitbhillai.ac.in OPT			
v Limit to display filter			
Flow type: All Flows			
Addresses: Any			

d. ARP Query

Filter used : arp

Time	Cisco_89:22:19	Intel_4a:52:4a	Cisco_af:3e:02	Cisco_0d:69:ff	Comment
35.607248		Who has 10.50.0.178? Tell 0.0.0.0			ARP: Who has 10.50.0.178? Tell 0.0.0.0
35.607276		10.50.0.178 is at 8c:e9:ee:4a:52:4a			ARP: 10.50.0.178 is at 8c:e9:ee:4a:52:4a
73.148733			Who has 10.50.0.1? Tell 10.50.0.178		ARP: Who has 10.50.0.1? Tell 10.50.0.178
73.152700			10.50.0.1 is at 34:1b:2d:af:3e:02		ARP: 10.50.0.1 is at 34:1b:2d:af:3e:02
113.052105			Who has 10.50.0.178? Tell 10.50.0.5		ARP: Who has 10.50.0.178? Tell 10.50.0.5
113.052131			10.50.0.178 is at 8c:e9:ee:4a:52:4a		ARP: 10.50.0.178 is at 8c:e9:ee:4a:52:4a
187.836658		Who has 10.50.0.1? Tell 10.50.0.178			ARP: Who has 10.50.0.1? Tell 10.50.0.178
187.839296		10.50.0.1 is at 34:1b:2d:af:3e:02			ARP: 10.50.0.1 is at 34:1b:2d:af:3e:02
341.331672		Who has 10.50.0.178? Tell 0.0.0.0			ARP: Who has 10.50.0.178? Tell 0.0.0.0
341.331696		10.50.0.178 is at 8c:e9:ee:4a:52:4a			ARP: 10.50.0.178 is at 8c:e9:ee:4a:52:4a
428.888878			Who has 10.50.0.178? Tell 10.50.0.5		ARP: Who has 10.50.0.178? Tell 10.50.0.5
428.888909			10.50.0.178 is at 8c:e9:ee:4a:52:4a		ARP: 10.50.0.178 is at 8c:e9:ee:4a:52:4a
578.492719		Who has 10.50.0.1? Tell 10.50.0.178			ARP: Who has 10.50.0.1? Tell 10.50.0.178
578.517265		10.50.0.1 is at 34:1b:2d:af:3e:02			ARP: 10.50.0.1 is at 34:1b:2d:af:3e:02
643.935035		Who has 10.50.0.178? Tell 0.0.0.0			ARP: Who has 10.50.0.178? Tell 0.0.0.0
643.935052		10.50.0.178 is at 8c:e9:ee:4a:52:4a			ARP: 10.50.0.178 is at 8c:e9:ee:4a:52:4a
649.660729			Who has 10.50.0.1? Tell 10.50.0.178		ARP: Who has 10.50.0.1? Tell 10.50.0.178
649.665068			10.50.0.1 is at 34:1b:2d:af:3e:02		ARP: 10.50.0.1 is at 34:1b:2d:af:3e:02
675.260725		Who has 10.50.0.1? Tell 10.50.0.178			ARP: Who has 10.50.0.1? Tell 10.50.0.178
675.283598		10.50.0.1 is at 34:1b:2d:af:3e:02			ARP: 10.50.0.1 is at 34:1b:2d:af:3e:02

4 nodes, 20 items

Limit to display filter

Flow type: All Flows

Addresses: Any

e. Pinging tesla.com

Filter used : ip.addr == 2.18.53.207

Time	10.50.0.178	2.18.53.207	Comment
519.123098	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
519.123118	Echo (ping) request id=0x3c0a, seq=1/256, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=1/256...
520.124902	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
520.124944	Echo (ping) request id=0x3c0a, seq=2/512, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=2/512...
521.148896	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
521.148956	Echo (ping) request id=0x3c0a, seq=3/768, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=3/768...
522.172619	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
522.172675	Echo (ping) request id=0x3c0a, seq=4/1024, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=4/1024...
523.196828	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
523.196885	Echo (ping) request id=0x3c0a, seq=5/1280, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=5/1280...
524.220936	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
524.221002	Echo (ping) request id=0x3c0a, seq=6/1536, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=6/1536...
525.244735	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
525.244793	Echo (ping) request id=0x3c0a, seq=7/1792, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=7/1792...
526.268834	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
526.268896	Echo (ping) request id=0x3c0a, seq=8/2048, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=8/2048...
527.292916	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
527.292963	Echo (ping) request id=0x3c0a, seq=9/2304, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=9/2304...
528.316924	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
528.316992	Echo (ping) request id=0x3c0a, seq=10/2560, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=10/256...
529.340853	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
529.340903	Echo (ping) request id=0x3c0a, seq=11/2816, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=11/281...
530.364885	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
530.364946	Echo (ping) request id=0x3c0a, seq=12/3072, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=12/307...
531.388890	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
531.388951	Echo (ping) request id=0x3c0a, seq=13/3328, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=13/332...
532.413818	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
532.413878	Echo (ping) request id=0x3c0a, seq=14/3584, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=14/358...
533.436804	Fragmented IP protocol (proto=ICMP 1, off=0, ID=c0)		IPv4: Fragmented IP protocol (proto=ICMP 1, off=...
533.436850	Echo (ping) request id=0x3c0a, seq=15/3840, ttl=64 (fr...		ICMP: Echo (ping) request id=0x3c0a, seq=15/384...

Packet 58855: ICMP: Echo (ping) request id=0x3c0a, seq=4/1024, ttl=64 (no response found)

Limit to display filter

Flow type: All Flows

Addresses: Any